

# CHEN (STEVEN) ZENG

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## PROFESSIONAL SUMMARY

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Full Stack Engineer with over 4 years of software development experience and an Electrical Engineering background. Focused on building AI Agentic workflows (Model Context Protocol), **Next.js**, and **React**. Proven track record of architecting scalable Educational Tech ecosystems at BCIT. Passionate about bridging the gap between LLMs and functional software tools to automate complex workflows.

## PROFESSIONAL EXPERIENCE

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### Multimedia Developer (Software Developer) | BCIT, Burnaby, BC | July 2023 – Present

- **Lead Developer & Architecture:** Architected the *Kombucha Guide*, a decoupled frontend framework that provided embeddable web apps within the D2L learning platform.
- **System Reliability:** Implemented End-to-End (E2E) testing suites using **Cypress** and unit testing with **Jest**, significantly reducing regression bugs in production environments.
- **API Integration:** Engineered a decoupled frontend that consumes data from the Kombucha Framework API via **SWR**, ensuring real-time synchronization of each web app's content.
- **CI/CD:** Established automated workflows for versioning and deployment, streamlining the release process for internal tools.

### Co-Founder & Lead Eng. | Top Gyu Cabin Corp., Richmond, BC | Dec 2021 – July 2023

- **Technical & Digital Infrastructure:** Built and managed the company's entire technical ecosystem from scratch, including the e-commerce web platform, corporate email systems, and domain administration.
- **Workflow Automation (React):** Engineered a custom internal dashboard using **React** and **Bootstrap** to track inventory and operational schedules, replacing manual processes for the founding team.
- **Business Operations & Compliance:** Established end-to-end B2B invoicing workflows and managed regulatory compliance (including complex health safety permitting), ensuring seamless financial operations and tax reporting.
- **Strategic Resource Management:** Directed financial tracking and resource allocation for the venture, bridging the gap between technical requirements and business constraints.

### Software Developer | Ensemble Systems, Vancouver, BC | Sep 2022 – Nov 2022

- **High-Traffic E-Commerce:** Translated Figma designs into pixel-perfect, responsive interfaces for **Best Buy Canada's** e-commerce platform using **React** and **TypeScript**.
- **State Management:** Integrated backend services via **GraphQL** and **PostgreSQL**, managing complex state flows with **Redux**.
- **Code Quality:** Participated in code reviews and implemented unit tests to strictly adhere to enterprise quality standards.

### Front-End Web Developer | Cine Gears Inc., Richmond, BC | July 2021 – Dec 2021

- **Dashboard & Plugins:** Built a real-time workforce dashboard using **React** and **Redux** and developed custom PHP plugins for WordPress to optimize e-commerce performance.
- **Feature Delivery:** Managed the end-to-end **SDLC** for client-facing web features, translating business requirements into responsive UI components.

## FEATURED PROJECTS

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### Weather MCP Server (AI Agent Tooling) | TypeScript, Zod, Model Context Protocol

- **MCP Server Development:** Developed a fully functional Model Context Protocol (MCP) server that exposes real-time weather data as executable tools for LLM agents (Claude/Cursor).
- **Type Safety & Validation:** Enforced strict type safety and input validation using Zod schemas to ensure reliable tool-calling behavior for stochastic AI models.
- **System Architecture:** Designed the architecture as a standalone Node.js executable, adhering to the standardized MCP client-host protocol to demonstrate readiness for AI-native infrastructure.

### AI-Driven Dev Workflow (Gemini CLI) | Gemini CLI, Google Antigravity

- **Generative UI Workflow:** Executed a “Generative UI” case study by using the Gemini CLI and MCP to scaffold and refactor a full-stack web application.
- **Automated Scaffolding:** Utilized natural language prompting to manipulate the file system, generate type-safe components, and implement state logic within an experimental JavaScript Framework environment.
- **AI-Augmented Testing:** Integrated Vitest for unit testing, demonstrating a robust AI-augmented development lifecycle.

### Full-Stack Portfolio Platform | Next.js, React, Tailwind CSS, Prisma

- **Modern CMS Architecture:** Engineered a high-performance CMS using the latest **Next.js App Router** and React Server Components.
- **Database Schema Design:** Designed a custom relational database schema and implemented the data model using **Prisma ORM** to ensure strict type safety.
- **Production Deployment:** Deployed the full-stack application to **Vercel** and integrated a serverless **Neon (PostgreSQL)** database to manage persistent user data.

## TECHNICAL SKILLS

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- **AI & Agentic Tech:** Model Context Protocol (MCP), LLM Tool Integration, Gemini CLI, Zod (Schema Validation).
- **Frontend:** React, Next.js (App Router), TypeScript, Tailwind CSS.
- **Backend:** Node.js, Express, PostgreSQL, Prisma ORM, MongoDB, GraphQL, REST APIs.
- **Testing & DevOps:** Cypress (E2E), Jest, Docker.
- **Tools:** Git/GitHub, GitLab, Visual Studio Code, Figma, Jira, Confluence, Postman

## EDUCATION

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### British Columbia Institute of Technology (BCIT), Burnaby, BC | 2020 – 2021

- Associate Certificate, Applied Web Development (GPA: 86% w/ Distinction)

### University of British Columbia (UBC), Vancouver, BC | 2011 – 2016

- Bachelor of Applied Science, Electrical Engineering